

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCIC602 - AIRPORT ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 **Which international organization is responsible for framing standards and recommended practices for civil aviation worldwide?**
A. FAA
B. IATA
C. ICAO
D. DGCA
- 2 **The primary objective of national air transport policy is to ensure-**
A. Growth of military aviation
B. Safe, efficient, and economical air transport services
C. Increase in aircraft manufacturing
D. Control of private airlines
- 3 **Which aircraft characteristic most directly influences runway and taxiway design?**
A. Seating capacity
B. Wing span
C. Cabin layout
D. Number of crew members
- 4 **The main advantage of regional airport planning is-**
A. Increase in runway length
B. Reduction in fuel consumption
C. Avoidance of duplication of airport facilities
D. Improvement in aircraft speed
- 5 **Wind data used for determining runway orientation is obtained from-**
A. Topographic map
B. Aeronautical chart
C. Wind rose diagram
D. Climatic report
- 6 **Airport zoning laws are mainly intended to-**
A. Control air traffic
B. Restrict aircraft noise
C. Prevent obstructions to safe aircraft operations
D. Increase land value
- 7 **The crosswind component on a runway should not exceed-**
A. 5 km/h
B. 10 km/h
C. Permissible limit of the design aircraft
D. Average wind speed
- 8 **Exit taxiways are provided in airports primarily to-**
A. Increase runway length
B. Reduce runway occupancy time
C. Park aircraft
D. Improve terminal access

- 9 **The LCN system in airfield pavement design is used to-**
 - A. Determine pavement thickness
 - B. Classify aircraft type
 - C. Compare pavement strength with aircraft loading
 - D. Identify pavement failures
- 10 **The basic function of an airport terminal building is to-**
 - A. Control aircraft movements
 - B. Provide parking for aircraft
 - C. Process passengers, baggage, and cargo efficiently
 - D. Store aviation fuel
- 11 **Airport grading is carried out mainly to-**
 - A. Improve aesthetics of the airport
 - B. Ensure proper surface drainage and operational safety
 - C. Reduce construction cost
 - D. Increase land value
- 12 **Airport lighting and visual aids are provided mainly to**
 - A. Beautify the airport
 - B. Assist pilots during night and low-visibility operations
 - C. Reduce fuel consumption
 - D. Improve terminal safety

Q2. Solve the following.

- A) **Explain the history and development of air transport and state its importance in present-day transportation.** 6
- B) **Describe the role and functions of air transport authorities in civil aviation.** 6

Q3. Solve the following.

- A) **Explain the concept of regional airport planning and discuss its advantages.** 6
- B) **Discuss the important factors considered in airport site selection and planning as per ICAO and FAA.** 6

Q4. Solve Any Two of the following.

- A) **With a neat sketch, explain runway orientation using wind rose and calculate wind coverage.** 6
- B) **Explain factors affecting runway length and illustrate corrections with a numerical example.** 6
- C) **With a sketch, explain taxiway layout and exit taxiway design.** 6

Q5. Solve Any Two of the following.

- A) **With a neat sketch, explain the functional layout of an airport terminal building.** 6
- B) **Explain airport grading and its importance in airport operations.** 6
- C) **Describe surface and subsurface drainage systems used at airports.** 6

Q6. Solve Any Two of the following.

- A) **Explain the need for air traffic control and describe the main functions of airfield pavement.** 6
- B) **Describe various air traffic control aids used at airports with common types of airfield pavement failures.** 6
- C) **Explain the need for maintenance and rehabilitation of airfield pavements.** 6

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